Exp 25 : RSA Private Key
Code:

```
#include <stdio.h>

int gcd(int a, int b) {
   while (b) {
      int t = b;
      b = a % b;
      a = t;
   }
   return a;
}

int main() {
   int n, c;

   printf("Enter n and ciphertext: ");
   scanf("%d %d", &n, &c);

   int g = gcd(n, c);
   printf("GCD = %d\n", g);

   if (g > 1 && g < n)
      printf("Factor found: %d\n", g);
   else
      printf("No nontrivial factor found\n");

   return 0;
}
```


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Language C
main.c
close ad [x]
#include <stdio.h>
int gcd(int a, int b) {
while (b) {
int t = b;
b = a % b;
a = t;
}
return a;
}
int main() {
int n, c;
printf("Enter n and ciphertext: ");
scanf("%d %d", &n, &c);
int g = gcd(n, c);
printf("GCD = %d\n", g);
if (g > 1 && g < n)
printf("Factor found: %d\n", g);
else
printf("No nontrivial factor found\n");
return 0;
}
input
Enter n and ciphertext: 3233 2790
GCD = 1
No nontrivial factor found
...Program finished with exit code 0
Press ENTER to exit console.